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Report

Laboratory Work No. 1

Operating Systems

Topic: The OS as a Resource Manager

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1 Purpose of the Laboratory Work

To observe directly from the terminal how the kernel of a Linux operating system acts as a resource manager interposed between user processes and the hardware components. The laboratory covers the four major responsibilities of an operating system: file management, process scheduling, memory administration, and the abstraction of input/output devices.

2 Initial System Information

```
$ whoami
adrian

$ uname -a
Linux adrian-QEMU-Virtual-Machine 7.0.0-38-generic #38-Ubuntu SMP PREEMPT_DYNAMIC
Fri Sep  4 09:28:02 UTC 2026 aarch64 GNU/Linux

$ uptime
 16:05:52 up 30 min,  2 users,  load average: 0.37, 0.24, 0.12
```

3 Part 1: File and Directory Management

3.1 Commands executed and relevant output

```
$ cd ~/os-lab1
$ mkdir demo && cd demo
$ echo "hello operating systems" > note.txt
$ cp note.txt copy.txt
$ mv copy.txt renamed.txt
$ ls -l
total 8
-rw-rw-r-- 1 adrian adrian 24 Oct  5 16:06 note.txt
-rw-rw-r-- 1 adrian adrian 24 Oct  5 16:06 renamed.txt

$ chmod 600 note.txt
$ ls -l note.txt
-rw------- 1 adrian adrian 24 Oct  5 16:06 note.txt
```

3.2 Observations and Answers (Part 1)

1. Owner and group: The created files (`note.txt`, `renamed.txt`) belong to the user `adrian` and the group `adrian` — on Ubuntu, every user receives by default a private group with the same name.

2. The meaning of the 10 characters after `chmod 600` (`-rw-------`):

- The first character (`-`) identifies the entry as a *regular file*.
- The next three characters (`rw-`) show the owner's permissions: read (`r`) and write (`w`), with no execute right (`-`).
- The next three characters (`---`) show that the members of the group have no permissions at all.
- The last three characters (`---`) show that all other users (*others*) have no permissions at all.

3. Two directories located directly under `/`:

- `/etc`: hosts the system-wide configuration files and directories.
- `/dev`: contains the special file nodes (*device files*) through which the kernel exposes physical and virtual hardware to user space.

4 Part 2: Process Management

4.1 Commands executed and relevant output

```
$ ps aux | head -5
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.3  26956 12780 ?        Ss   15:35   0:01 /usr/lib/systemd/systemd
                                                                    --switched-root --system
                                                                    --deserialize=53 splash
root           2  0.0  0.0      0     0 ?        S    15:35   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    15:35   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   15:35   0:00 [kworker/R-rcu_gp]

$ ps aux | wc -l
226

$ sleep 300 &
[1] 7568
$ cat /proc/7568/status | head -3
Name:   sleep
Umask:  0002
State:  S (sleeping)
```

4.2 Observations and Answers (Part 2)

1. Identifying the process with PID 1: The process is `/usr/lib/systemd/systemd` (traditionally reachable through the `/sbin/init` link). It is the initialization process launched by the kernel at boot and the direct or indirect parent of every process in user space.

2. Number of active processes on the VM: The command `ps aux | wc -l` reported 226 lines, i.e. 225 processes plus the header line.

3. The state of the process read from `/proc/<PID>/status`: The line shows `State: S (sleeping)`, the *interruptible sleep* state in which the process waits passively for its timer to expire or for a signal to arrive, without consuming any CPU time.

5 Part 3: Memory Management

5.1 Commands executed and relevant output

```
$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.3Gi       1.9Gi       126Mi        48Mi       1.4Gi       1.4Gi
Swap:          3.8Gi       566Mi       3.2Gi

$ grep VmRSS /proc/7568/status
VmRSS:      6352 kB
```

5.2 Observations and Answers (Part 3)

1. Total and free RAM: The virtual machine has a total of **3.3 GiB** of RAM, of which **126 MiB** are completely unallocated (*free*), while **1.4 GiB** are reported as effectively usable (*available*) thanks to the reclaimable page cache (*buff/cache*).

2. Definition of swap memory and the configured amount: Swap is an area on disk used as an extension of physical memory for inactive pages; when RAM runs low, the kernel moves such pages out to it. The system has **3.8 GiB** of swap space configured, of which 566 MiB are currently in use.

3. Resident memory (VmRSS) of the `sleep` process: The value is **6352 kB** (≈6 MiB). Such a small footprint is expected, since the `sleep` binary only loads the C standard library (libc) and directly invokes a pausing system call without allocating any additional data structures.

6 Part 4: Device and Storage Management

6.1 Commands executed and relevant output

```
$ lsblk
NAME   MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0    7:0    0     4K  1 loop /snap/bare/5
loop1    7:1    0  19.9M  1 loop /snap/desktop-security-center/152
loop2    7:2    0  61.9M  1 loop /snap/core24/1644
loop3    7:3    0 247.5M  1 loop /snap/firefox/8762
...
sr0     11:0    1   3.9G  1 rom  /run/media/adrian/Ubuntu 26.04.1 LTS arm64
vda    253:0    0    64G  0 disk
|-vda1 253:1    0     1G  0 part /boot/efi
`-vda2 253:2    0  62.9G  0 part /

$ df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/vda2        62G   11G   48G  19% /

$ ls -l /dev | head -4
total 0
crw-r--r--  1 root   root     10, 235 Oct  5 15:35 autofs
drwxr-xr-x  2 root   root        360 Oct  5 15:36 block
drwxr-xr-x  2 root   root         60 Oct  5 15:35 bsg
```

6.2 Observations and Answers (Part 4)

1. The device on which `/` is mounted: The root file system is mounted on the disk partition `/dev/vda2` — the second partition of the VirtIO virtual disk `vda` (64 GB), the first partition `vda1` being reserved for the EFI boot area.

2. An example entry from `/dev`: The block special file `sr0` (associated with the `cdrom` symbolic link) represents the optical CD/DVD-ROM unit, on which the Ubuntu 26.04.1 LTS installation image is currently mounted. A character-device example is `autofs` (first character `c`), used by the kernel's automounter.

3. The meaning of the “everything is a file” concept: This UNIX-specific paradigm abstracts hardware devices, communication channels, and kernel structures as files within a single unified tree, allowing them to be manipulated uniformly through the classic system calls: `open`, `read`, `write`, and `close`.

7 Final Synthesis

The operating system fulfils its role of resource manager by efficiently structuring **files** into an accessible hierarchy, explored with the `ls` command; by scheduling and isolating **processes** on the CPU cores, actively monitored through the `ps` and `top` utilities; by dynamically distributing **RAM** and swap space, evaluated with the `free` command; and by the unified abstraction of hardware and storage **devices**, identified in the system through the `lsblk` command, in accordance with the UNIX philosophy.